

DUSTBADGE · REQUIREMENTS

DustBadge requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/dustbadge		
Source	docs/03-requirements.md @ 0bdd5bb		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-25	Amish Chadha	Initial scaffold

DustBadge requirements

These are first-pass requirements for the concept. Targets are proposals for review and will be checked by calculation at TRL 3. Two requirements are not met by the concept as drawn (R2 and R14), and several others are unverified; the status column says which.

Table 1. Requirements and concept status.

ID	Requirement	Target	Status at TRL 2 (estimate)	Verification (TRL 3 or later)
R1	Measure the respirable dust fraction	Mass concentration of particles up to 4 µm (PM4), a proxy for the ISO 7708 respirable convention	Met by design with an SPS30-class sensor; optical sizing is not aerodynamic sizing, so the match is approximate	Datasheet review; later comparison with a cyclone sampler
R2	Cover the working range	0 to 5 mg/m ³ respirable dust without saturating	Not met. The proposed sensor is specified to 1 mg/m ³ ; readings above are uncalibrated	Calculation of expected site ranges; chamber test on CalRig
R3	Accuracy after site calibration	Shift average within ±25 % of a co-located gravimetric sample	Unverified; NIOSH reports factory-calibrated optical monitors can differ by up to 10 times without a site factor	Error budget at TRL 3; later co-located field samples
R4	Estimate silica exposure	RCS estimate = calibrated PM4 mass x site silica fraction, always labeled as an estimate; dust shown alone when no fraction is entered	Met by design	Firmware sketch review
R5	Update often enough to catch tasks	1 s readings, 1 min logged averages, running 8 h time-weighted average (TWA) updated every minute	Met by design	Firmware sketch review
R6	Warn before a limit is reached	Vibration and LED when the projected 8 h RCS TWA exceeds the action level (default 25 µg/m ³) and a repeated alert at the limit (default 50 µg/m ³); both configurable	Met by design	Firmware sketch review
R7	Last a full shift	12 h of continuous sampling per charge	At risk: about 13 h typical, about 11 h at the sensor's maximum current	Power budget calculation
R8	Wearable in the breathing zone	Worn within 30 cm of the nose and mouth; clip or harness loop	Met: chest mount about 22 cm below the nose in the massing model	Massing model
R9	Light and small	Mass 120 g or less; no larger than 75 x 55 x 35 mm	Met, thin margin on mass: about 110 g, 64 x 52 x 30 mm	Massing model, then weighing
R10	Survive the site	Electronics splash and dust protected (IP54 target except the sensor air path); inlet facing down; survives a 1.5 m drop onto concrete	Unverified	Design review; later drop and spray tests
R11	Work in site conditions	0 to 45 °C, 10 to 90 % RH non-condensing; readings flagged when humidity or spray may bias them	Unverified; water mist inflates optical readings	Literature and datasheet review
R12	Keep a shift log	At least 30 shifts of 1 min records on the badge; export over Bluetooth Low Energy to the worker's phone	Met: about 12 kB per 12 h shift	Storage calculation

ID	Requirement	Target	Status at TRL 2 (estimate)	Verification (TRL 3 or later)
R13	Protect the worker's data	Stored on the badge and the worker's phone; shared with an employer only by the worker's choice	Met by design	Design review
R14	Safe in hazardous atmospheres	Certified intrinsically safe for gassy mines and explosive atmospheres	Not met and out of scope for this prototype; it must be labeled not for such use	Design review
R15	Low cost and buildable	Parts \$90 or less; no custom PCB required for the first build	Met: about \$85 (indicative); a simple carrier board may be hand-wired on perfboard	Priced BOM

■ Assumptions

- The respirable convention in ISO 7708 has a 50 % cut near 4 μm aerodynamic diameter; PM4 from an optical sensor is used as a proxy.
- A site silica fraction (the quartz share of respirable dust) is available for each site and task from at least one filter sample analyzed by a laboratory or a field infrared method.
- Default limits follow US OSHA and MSHA (50 $\mu\text{g}/\text{m}^3$ limit and 25 $\mu\text{g}/\text{m}^3$ action level, 8 h TWA). Other jurisdictions can be configured, for example the EU binding limit of 100 $\mu\text{g}/\text{m}^3$.
- Shifts are up to 12 h; the projected TWA is normalized to 8 h as the regulations define it.